Bharath Kumar T | Software Quality Engineer |Validation Engineer

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PROFESSIONAL SUMMARY

- 5+ years of experience in automotive embedded systems testing, OTA flashing, HIL/SIL validation, and system-level integration testing.
- Skilled in test case design, execution, and management using TestRail, JIRA, Zephyr, ensuring requirement, regression, and defect tracking.
- Expertise in AUTOSAR (ARXML), ISO 26262, and vehicle protocols: CAN, CAN FD, LIN, Ethernet, FlexRay, OBD-II, UDS, DoIP.
- Proficient in automation and DevOps using Python, Pytest, Robot Framework, REST API/Swagger testing, and CI/CD pipelines via Jenkins, Git/GitHub, Bitbucket, Docker, Kubernetes, Azure DevOps.
- Strong ability in log analysis, OTA monitoring, dashboards, and cross-ECU validation, improving defect detection, test coverage, and release quality by 40%.

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, SQL, MATLAB, SIMULINK
- **Testing & Quality Assurance:** Manual Testing, Automation Testing, Test Case Design, Regression Testing, System-Level Testing, TestRail, JIRA, Jenkins, QNX, VEX, JFrog, LabView
- **Automation Frameworks:** TestNG, JUnit, Pytest, Appium, Robot Framework
- **Methodologies & Practices:** Agile/Scrum, BDD, TDD, SDLC, STLC, Data-Driven Testing, Smoke Testing
- **Embedded & Vehicle Testing:** Vector CANoe, dSpace, vSpy, DPS Tool, CAPL scripting, CAN/LIN/MOST/Ethernet protocols, OTA updates, VHAL
- **Mobile & Cross-Platform Testing:** Android Auto, Apple CarPlay, iOS & Android (Appium, native tools), cross-browser testing
- Databases & ORM: Oracle, MySQL, SQL Server, PostgreSQL, PL/SQL
- **Cloud & Platforms:** AWS (EC2, S3, Lambda, RDS, CloudFormation)
- **IDEs & Development Tools:** Eclipse, IntelliJ IDEA, Visual Studio, Visual Studio Code
- **CI/CD & Version Control:** Jenkins, Docker, Kubernetes, Git, Bitbucket.

PROFESSIONAL EXPERIENCE

Software Engineer

Sirius XM (Farmington Hills, Michigan, USA) June 2025 - Present

- Tested SXM software modules using TestRail and JIRA, executing 80–100+ test cases per release with >95% coverage; developed automation scripts in Python, Pytest, Robot Framework, improving regression efficiency by ~35%.
- Executed and validated Over-the-Air (OTA) software updates, including deployment, installation verification, rollback, and monitoring update status, reducing post-release software issues by ~20%.
- Developed and executed HIL/SIL test cases for system-level validation, detecting defects ~30% earlier in the development lifecycle, reducing dependency on manual testing.
- Managed test cases, execution, and defect tracking using TestRail and JIRA, ensuring 100% requirement traceability and improving defect resolution time by ~25–30%.
- Supported CI/CD and DevOps processes using Jenkins, Git/GitHub, Docker, and Azure DevOps, automating builds, test execution, and reporting, accelerating release cycles by ~30%.
- Created automated dashboards, logs, and coverage reports, enhancing visibility of testing progress and enabling faster cross-team collaboration, improving early defect detection by ~25%.
- Developed Python-based tools for log parsing, test result analysis, and report generation, reducing manual analysis effort by ~50% and improving defect triage speed.
- Supported integration testing with backend services and cloud-based systems, leveraging REST API and Swagger/OpenAPI for automated validation of system interactions.

Software Engineer

General Motors(Warren, Michigan, USA) February 2024 - May 2025

- Executed comprehensive test cases across vehicle modules including Instrument Cluster Display, Android, infotainment, HMI, ADAS, connectivity systems, and CS Display, ensuring 100% validation against functional, FMVSS safety standards, and design requirements.
- Conducted various testing types (monkey, smoke, sanity, functional, regression, acceptance, and end-to-end testing) to ensure system stability and performance, reducing post-release defects by 25%; applied knowledge of Ethernet protocols (TCP/IP, UDP, DoIP).

- Performed in-vehicle testing and software validation using MATLAB/Simulink and Vehicle Hardware Abstraction Layer (VHAL) to simulate real-world EV scenarios, improving integration efficiency by 30%.
- Supported software deployments by coordinating build releases and executing deployment scripts via JFrog Artifactory and GitHub, ensuring seamless integration in embedded automotive systems with zero rollback incidents over 12 months.
- Executed manual and Over-The-Air (OTA) ECU flashing to deploy software updates and verify system behavior post-flash; contributed to reducing on-road testing cycles by 20% through pre-validation in HIL/SIL environments.
- Utilized vSPY, NeoVI, and Vector CANoe for communication troubleshooting and system validation, extensively working with CAN 2.0, CAN FD, and OBD-II protocols; resolved 95% of communication issues during integration phase.
- Automated test scripts, MATLAB/Simulink, supported debugging using TeraTerm, simulators, and embedded platforms such as QNX and QFIL, cutting manual testing effort by 40%.
- Completed planned test cycles on schedule, executing ≥90% of test cases within committed timelines, ensuring adherence to delivery milestones and project goals.
- Applied knowledge of safety-critical systems, FMVSS compliance, and automotive standards (ISO 26262, AUTOSAR architecture) in all testing and validation activities.
- Performed validation using HIL and SIL environments to test software behavior under real-time simulation, reducing on-road testing dependencies by 25%.
- Identified, raised, and tracked software defects using JIRA and TestRail; performed live CAN log debugging and adb commands, reducing defect resolution time by 20%.
- Collaborated with cross-functional teams (developers, systems engineers, QA, product managers, UI/UX designers using Figma) to validate design, functionality, and performance of EV systems.
- Experienced in CI/CD pipelines and containerization tools (Docker, Kubernetes) to streamline software build, test, and deployment workflows, achieving a 15% faster release cycle.

Sodexo- Operations Team

(Kansas City, USA) August 2023 - December 2023

- Developed Python scripts to automate data tracking for dining, improving efficiency in inventory and order management.
- Used Python and Pandas to analyze daily dining data, identifying trends and optimizing workflow to reduce waste.
- Created automation tools to generate daily and weekly reports on stock levels, sales, and staff schedules.
- Built Python dashboards using Matplotlib and Seaborn to visualize trends in food consumption and inventory usage.
- Automated basic data entry and cleaning processes using Python scripts, minimizing errors and improving accuracy.
- Learned and applied Python libraries and frameworks (NumPy, Pandas, Matplotlib) to solve real-world operational problems.

Associate Software Engineer

Tata Consultancy Services (Bangalore, India) August 2020 - July 2022

- Developed 50+ Python scripts for automation of vehicle system testing, reducing manual effort by 40%.
- Designed and maintained Python-based automation frameworks using Selenium, PyAutoGUI, and Robot Framework, improving test efficiency by 35%.
- Built Python scripts for data extraction, transformation, and reporting on 10+ vehicle projects, delivering error-free results 50% faster.
- Created Python scripts integrated with APIs and web scraping tools (Requests, BeautifulSoup) to automate data collection, reducing manual work by 60%.

EDUCATION

Master of Science, Computer Science
University of Missouri, Kansas City August 2022 - December 2023
Bachelor of Technology, Electronics and Communication Engineering
Sri Venkateswara University, Tirupati July 2017 - July 2021

CERTIFICATIONS

- AWS Certified Developer - Associate
- Embedded System Design With ARM